

— Research Interests

Information retrieval, reinforcement learning, recommender systems, large language models, generative models and scaling laws, and the mathematical foundations of deep learning.

— Education

- 2024 – 2026 **M.Sc., Applied Mathematics and Computer Science, Moscow State University**
Faculty of Computational Mathematics and Cybernetics. Diploma with Honours.
Thesis topic: Approximation of softmax activation function in recommendation problem.
- 2020 – 2024 **B.Sc., Applied Mathematics and Computer Science, Moscow State University**
Faculty of Computational Mathematics and Cybernetics. Diploma with Honours.
Thesis topic: Application of discrete mathematics and combinatorics in machine learning.

— Work Experience

- 2026 – Present **Senior Researcher, AI VK**
Part of a research team led by [Prof. Dr. Alexander D'yakonov](#), formerly world's #1 rank on Kaggle.
Research topics: generative retrieval and reinforcement learning in recommender systems.
- Co-authored [Long-Term Optimization for Large-Scale Generative Retrieval with Off-Policy REINFORCE](#), accepted to the 5th Workshop on End-to-End Customer Journey Optimization at KDD 2026.
- 2024 – 2026 **Senior Deep Learning Engineer, RecSys R&D, Yandex**
Worked under the leadership of [Kirill Khrylchenko](#). Research topics: generative models and scaling laws, and linear attention in real-time recommendation models.
- Part of the team that developed [Argus](#). Argus increased TLT at Yandex Music by +2.26% and like likelihood by +6.37%. It was later adapted by our colleagues to nearly every recommendation-based Yandex product (e.g., e-commerce +1.7% GMV, food-tech services +2.1% GMV, and others).
 - Worked on quality-complexity trade-offs for recommendation models, co-authoring [Gated Bidirectional Linear Attention for Generative Retrieval](#).
 - Worked on pretraining large-scale generative recommender models.
- 2024 **Research Intern, RecSys R&D, Yandex**
Research topic: optimal ways to represent items in web-scale transformer-based recommenders. Authored [Embedding Items at Scale: Comparing GNN-Based and ID-Based Item Embeddings in the Yandex Ecosystem](#).
- 2021 – 2024 **Research Assistant, Moscow State University**
Research topics: cognitive dynamics analysis and internal speech recognition under Russian Science Foundation grants [#20-18-00067](#) and [#20-18-00067-II](#), under the supervision of [Dr. Alexander Vartanov](#).

— Publications

- KDD 2026** [Scaling Recommender Transformers to One Billion Parameters](#)
Kirill Khrylchenko, Artem Matveev, **Sergei Makeev**, Vladimir Baikalov
[Long-Term Optimization for Large-Scale Generative Retrieval with Off-Policy REINFORCE](#), *5th Workshop on End-to-End Customer Journey Optimization*
Artem Matveev, **Sergei Makeev**, Aleksei Krasilnikov, Vladimir Baikalov, Sergei Liamaev, Kirill Khrylchenko
- SIGIR 2026** [Gated Bidirectional Linear Attention for Generative Retrieval](#)
Artem Matveev, Vladislav Tytskiy, **Sergei Makeev**, Sergei Liamaev

RecSys 2025 [Blending Sequential Embeddings, Graphs, and Engineered Features: 4th Place Solution in RecSys Challenge 2025](#)
Sergei Makeev, Alexandr Andreev, Vladimir Baikalov, Vladislav Tytskiy, Aleksei Krasilnikov, Kirill Khrylchenko
[Correcting the LogQ Correction: Revisiting Sampled Softmax for Large-Scale Retrieval](#)
Kirill Khrylchenko, Vladimir Baikalov, **Sergei Makeev**, Artem Matveev, Sergei Liamaev

— Professional Service

CIKM 2026 PC member, Demo Track.
UMAP 2026 PC member, Industry Track.
UMAP 2025 PC member, Industry & Start-up Track.

— Patents

Dec 2025 Application for invention [2025136959](#), Federal Service for Intellectual Property
Yandex filed a patent application for Argus (see our paper [Scaling Recommender Transformers to One Billion Parameters](#)).

— Competitions & Olympiads

Jun 2025 RecSys Challenge 2025 (RecSys'25)
Led team **ambitious** to **4th place**.
Apr 2024 Moscow State University, [The Lomonosov Universiade](#) in Mathematics and Computer Science
2nd-degree diploma.
Apr 2023 Moscow State University, [The Lomonosov Universiade](#) in Mathematics and Computer Science
1st-degree diploma.

— Talks

Sep 2025 [Talk at RecSys'25](#), O2 Universum, Prague, Czech Republic
Presented my team's solution to the RecSys Challenge 2025 as an invited speaker.
Mar 2026 [Talk at HSE University](#), Moscow, Russia
Gave a guest lecture for the [DeepRecSys course](#) taught by Kirill Khrylchenko at the Faculty of Computer Science, HSE University.

— Teaching

- Contributed to the [DeepRecSys undergraduate course](#) taught by Kirill Khrylchenko at the Faculty of Computer Science, HSE University.

— Research in News and Social Media

- [Argus](#) received broad attention after its launch. It was featured in news outlets (e.g., [ict.moscow](#), [yagla.ru](#), [mltimes.ai](#), [setka.ru](#), [toolarium.ru](#), [m.seonews.ru](#), [rozetked.me](#), [pikabu.ru](#)), covered on [habr.com](#), and discussed on both [x.com](#) and [habr.com](#). It was later adapted by our colleagues to many other Yandex products. Its deployment in Yandex Ads also drew extensive discussion (e.g., [timeweb.com](#), [habr.com](#), [adindex.ru](#), [dyvannyi-it.ru](#), [marketplacegu.ru](#), [sostav.ru](#), etc.). [Sumit Kumar](#), author of a popular information retrieval blog (4.1k subscribers on x.com as of Jul 2026), wrote about Argus on [x.com](#) and listed it in [Top Information Retrieval Papers of the Week](#).
- Our paper [Correcting the LogQ Correction: Revisiting Sampled Softmax for Large-Scale Retrieval](#) also received extensive discussion: [vc.ru](#), [ict.moscow](#), [kommersant.ru](#), [naked-science.ru](#), [cnews.ru](#), [itsnew.ru](#), etc. It was listed among [Top Information Retrieval Papers of the Week](#).
- [Gated Bidirectional Linear Attention for Generative Retrieval](#), which I co-authored, was also noted by the community on [linkedin.com](#) and [x.com](#). It was listed in Sumit Kumar's popular blog [Top Information Retrieval Papers of the Week](#).
- [Long-Term Optimization for Large-Scale Generative Retrieval with Off-Policy REINFORCE](#) was also noted by Sumit Kumar on [x.com](#) and included in his [Top Information Retrieval Papers of the Week](#) rating. After its release, it was promptly cited by some scientific websites (e.g., [gist.science](#), [moltbook.com](#), [chatpaper.com](#)).

— Mentorship

- Mentored two interns at Yandex, Oleg Sorokin and Vladislav Dodonov. Both are now core developers on the RecSys R&D team (see, for example, [Gryphon](#)).

— Grants

- Worked on cognitive dynamics analysis and internal speech recognition under Russian Science Foundation grants [#20-18-00067](#) and [#20-18-00067-II](#) as a research assistant at Moscow State University, under the supervision of [Dr. Alexander Vartanov](#).

— Technical Skills

Languages Current: Python. Past: C/C++, Assembly, CUDA.
Frameworks Current: PyTorch. Past: TensorFlow, JAX.